Digitizing Midwifery Care: AR and VR Solutions for Maternal Education

25-26J-426

Project Proposal Report

Hasara M. A. H – IT22262936

BSc (Hons) in Information Technology – Interactive Media

Department of Information Technology

Sri Lankan Institute of Information Technology

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Supervisor - Mr. Didula Thanaweera Araghchi

Co-Supervisor - Mrs. Narmada Gamage

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Declaration page of the candidates & supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	**Student ID**	**Signature**
Hasara M. A. H	IT2262936	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for their undergraduate dissertations under my supervision.

Mr. Didula Thanaweera Arachchi

Signature of the supervisor: Date: 07/09/2024

Abstract

Midwives play a pivotal role in early childhood health across Sri Lanka, especially in rural and underserved communities. However, they often lack access to digital tools that help identify infant cranial and postural abnormalities early. This research proposes the development of a bilingual (Sinhala/English) mobile application that integrates an AI-assisted infant cranial and posture monitoring module. Using computer vision techniques, the app will analyze infant images or videos to detect cranial deformities and postural irregularities, providing actionable recommendations for early intervention. A bilingual interface will ensure accessibility for Midwives across diverse literacy and language backgrounds. The project aims to evaluate both the technical accuracy of the AI module and the usability of the bilingual interface with midwives in Sri Lanka.

Table of Contents

Declaration page of the candidates & supervisor iii
Abstract iv
Table of contents v
List of Figures vii
List of Tables viii
List of abbreviations ix
Introduction 1
Background & Literature survey 1
Research Gap 4
Table 1 Comparative Table 5
Research Problem 6
Objectives 7
Main Objectives 7
Specific Objectives 8
Figure 1 System Diagram 9
Methodology 10
Functional Requirements 11
Non-Functional Requirements 13
System Requirements 14
Hardware Requirements 14
Software Requirements 14
Table 2 Technologies Used 15
Figure 2 Gantt Chart 16
Figure 2 Work Breakdown Structure (WBS) 17
Description of Personal and Facilities 18
Supervisor 18
External Supervisor 18
Facilities 18

Device Access 18
Budget and Budget Justification 19
Table 3 Budget Table 19
Reference List 20

List of Figures

Figures 1 System Diagram 9

Figures 2 Gantt Chart 16

Figures 2 Work Breakdown Structure (WBS) 17

List of Tables

Table 1 Comparative Table 5

Table 2 Technologies Used 15

Table 3 Budget Table 19

List of abbreviations

AI	Artificial Intelligence
CNN	Convolutional Neural Network
CVAI	Cranial Vault Asymmetry Index
CI	Cephalic Index
mHealth	Mobile Health
UI	User Interface
UX	User Experience
SUS	System Usability Scale
IDE	Integrated Development Environment
API	Application Programming Interface
SQLite	Structured Query Language Lite
SDK	Software Development Kit
BaaS	Backend as a Service
FOSS	Free and Open-Source Software
LKR	Sri Lankan Rupees
NLP	Natural Language Processing
RGB	Red, Green, Blue (color model used in image processing)

ROI	Region of Interest (used in image analysis)
FPS	Frames Per Second
UI/UX	User Interface / User Experience
ML	Machine Learning
API	Application Programming Interface

Introduction

This research project will create and test a mobile phone app that helps midwives in Sri Lanka find early signs of head shape problems and body posture issues in babies. The app will use computer tools called TensorFlow and OpenCV to look at pictures or videos of infants and give important health numbers, such as CVAI (which shows if the head is flat on one side), CI (which shows the head shape), and posture angles (which show how the baby's body is positioned). The app will be built using Flutter and will use Firebase to store and manage data safely. It will support both the Sinhala and English languages so that midwives from different areas can use it easily, even if they speak different languages or have different reading skills. The project will begin with reading many research papers and preparing a dataset of baby images. Then, the team will build the AI system, design the mobile app, and connect the AI to the app so it can work in real time or with saved images. Midwives will test the app and give feedback using a score system called the System Usability Scale (SUS), and they will also share their thoughts to help improve the design. The AI system will be tested to see how correct and reliable it is by comparing it with expert standards. The final goal is to make a tool that fits the culture and needs of Sri Lankan midwives, helps them give better care, and improves the health of babies.

Background & Literature survey

In Sri Lanka, midwives are the main caregivers for mothers and babies, especially in villages and places with fewer resources. They play a very important role in keeping mothers and infants healthy. But many midwives do not have digital tools to help them find early signs of health problems in babies. Some babies may have head shape problems like plagiocephaly (flat head on one side) or brachycephaly (short and wide head), or body posture issues. If these problems are not found early, they can cause long-term health issues. It is very important to find these problems early and give care

quickly. But right now, most ways to check babies are done by hand, depend on personal judgment, or require special machines that are not available in poor areas. This research project wants to solve that problem by making a mobile app that uses AI (artificial intelligence) to help midwives check babies' head shapes and body posture. The app will work in both Sinhala and English so that midwives with different reading and language skills can use it easily.

New mobile health technologies have shown that it is possible to use computer vision and machine learning to study baby pictures and give useful health information. For example, Nakahari and others tested an app called "Measurement of Baby's Head Shape™." This app uses smartphone pictures to calculate CVAI (Cranial Vault Asymmetry Index) and CI (Cephalic Index), which are numbers that show head shape. The app worked very well, with high accuracy and good results. But it was only tested on plastic head models, not on real babies. So, it still needs more testing in real-life situations, where babies move, lighting changes, and hair may cover parts of the head.

Another study by Neves and others created a tool called PosturAll. It looks at still pictures and gives scores to show the risk of body posture problems. It worked well and gave results **like** professional tools. But it was tested only on adult students, not on children. So, it may not work the same way for babies. Still, the tool works on mobile phones and uses smart ways to find body markers, which means it could be changed to help with child health.

Tabatabaei and others made an app called INCROS. It uses deep learning to find head shape problems from smartphone pictures. The app uses a smart model called U-Net with ResNet-18 and gives results in just 11 seconds. It worked very well, with 99.01% accuracy. The app is easy to use caregivers only need a tight cap and a coin for size reference. But the app was tested on a small number of pictures, and it still needs to be tested in real-world settings.

Some researchers have also used AI to study signals from the baby's body. León and others made a model that uses heart rate, breathing, and other signals to guess the baby's

developmental age. The model was very accurate and helped find delays in very small or early born babies. This shows that AI can help with baby health in ways that do not need touching or special machines.

Shamsir and others went even further. They made a system with many sensors and AI to find problems like breathing stops (apnea), seizures, and discomfort. It worked very well with fake data and gave alerts in real time using mobile and cloud systems. But because it used fake data and was not tested in hospitals, it cannot be used in clinics yet. Still, it shows how sensors and AI can work together to care for babies.

In the area of mothers' health, wearable tools and apps that use two languages have helped people make better health choices. Zhang and others made a wearable system that uses AI to study labor progress and decide when a mother should go to the hospital. It worked well and was tested against hospital machines. But it only worked in one language, so it was hard for people who speak other languages to use it.

AI has also been used in talking systems. Al-Mutairi and others made a bilingual chatbot that helps with school advice in English and Arabic. It was not made for health, but it's designed to separate AI systems for each language, and as a backup system when the answer is not clear, it can be used in health apps too. But the Arabic part had problems like no spell check and weak grammar tools, which show that language tools need to be better for health use.

All these studies show that AI-powered mobile apps can change how we care for mothers and babies. They show that it is important to make tools that are easy to use, accurate, and available to everyone, especially in places with fewer resources. Your research project builds on these ideas. You are making a bilingual mobile app for midwives in Sri Lanka. It will use computer vision to study baby head shapes and posture. The app will be designed for people with different reading levels and language skills. This mix of smart technology and kind design can help midwives give better care and improve the health of babies across the country.

Research Gap

One big problem in checking baby health is that there are not enough easy-to-use and trusted tools to find head shape problems in real-life places like homes or small clinics. Some apps, like Measurement of Baby's Head Shape™ (by Nakahari) and INCROS (by Tabatabaei), show good results in finding head shape issues like plagiocephaly (flat head on one side) and brachycephaly (short and wide head). But these apps were tested in special, controlled places. Nakahari's app was tested only on plastic head models, not real babies. INCROS needed a tight cap and a coin to help measure the head correctly. Even though INCROS gave very accurate results—more than 99%, it was only tested on 28 baby pictures. These limits make it hard to trust apps in real-life places like village clinics or homes. To solve this, my research will create a mobile app that works in both Sinhala and English. It will use AI (artificial intelligence) to find head shape problems using real baby pictures taken in everyday settings. The app will be tested with midwives in Sri Lanka to make sure it works well and is easy to use in places with fewer resources. Another big gap is that there are no good tools made just for checking baby posture. One tool called PosturAll (by Neves) uses pictures and smart computer methods to check body posture. It gave good results—over 92% accuracy. But it was only tested on adult students, not babies. Babies move and grow differently from adults, so the tool may not work well for them. My research will add posture checking to the mobile app using pictures and data made just for babies. The AI system will be trained using baby-specific images. Midwives will test the app to make sure it fits well into their daily work and helps them care for babies better.

Language is also a big barrier in health technology, especially in places like Sri Lanka, where people speak different languages. One example is a bilingual chatbot made by Al-Mutairi. It used English and Arabic and showed that it is possible to build systems that understand more than one language. But this chatbot was made for school advice, not health care. It only gave 75-80% correct answers, which is not good enough for health use. Also, the Arabic part had problems like no spell-check and weak grammar tools,

which can be risky in health settings. My research will create a Sinhala and English interface made specially for midwives. It will use clear health words and smart language tools to answer common questions and give guidance. If the app is not sure about a question, it will show helpful pictures or messages to keep things safe and easy to understand.

Lastly, there are very few tools that can watch both mothers and babies in real time and are also easy to use for different cultures and languages. Some systems, like the wearable labor monitor (by Zhang) and the smart baby monitor (by Shamsir), use sensors and AI to check body signals and send alerts to phones. These systems worked well—over 96% and 99.88% accurately—but they were tested only on small groups or fake data. They did not support more than one language, and their designs were not made for people from different backgrounds. They also did not have chat support to help users. My research will fill this gap by making a mobile app that checks a baby's head shape and posture using AI and includes a bilingual chat system. The app will protect privacy, give alerts in real time, and offer advice that fits the local culture. This will help midwives and families in Sri Lanka take better care of babies using one simple, inclusive digital tool.

Table 1 Comparative Table

Area	**Existing Studies / Tools**	**Strengths**	**Gaps**
Head Shape Analysis	*Measurement of Baby's Head Shape*™ (Nakahari) *INCROS* (Tabatabaei)	- Automated CVAI & CI calculation - High accuracy (>99%)	- Tested on models or small datasets - Needed caps/coins - Lab conditions only
Posture Analysis	*PosturAll* (Neves)	- Accurate posture risk scoring (>92%)	- Tested on adults, not babies

Real-World Testing	Multiple studies	- Showed proof of concept	- No field testing with health workers
Bilingual Support	*Al-Mutairi's Chatbot*	- Multi-language design	- Low accuracy (75–80%) - No health focus
Mother & Baby Monitoring	*Labor Monitor* (Zhang) *Smart Baby Monitor* (Shamsir)	- High accuracy, real-time alerts	- Small samples, simulated data - No language/cultural support

Research Problem

In Sri Lanka, midwives are very important for checking the health and growth of babies, especially in villages and places with fewer resources. But many midwives do not have digital tools to help them find early signs of problems in a baby's head shape or body posture. These problems, like flat head or poor posture, can cause serious health issues later if they are not found and treated early. Right now, most ways to check these problems are done by hand, depending on personal judgment, or require special machines that are not available in small clinics or homes. Also, many health apps do not support the Sinhala language, so Sinhala-speaking midwives cannot use them easily. This research aims to solve both problems: the lack of accurate tools and the lack of language support. It will create a mobile app that works in both Sinhala and English. The app will use AI (artificial intelligence) to study pictures and videos of babies and find head shape and posture problems. It will use computer vision to do this and show the results in a simple and culturally friendly way. By mixing smart technology with kind design, this project hopes to give midwives a tool that is both medically correct and easy to use, helping them take better care of babies all across Sri Lanka.

Objectives

This research project has clear goals. It wants to help midwives in Sri Lanka by giving them a smart and easy-to-use tool to find early signs of problems in a baby's head shape and body posture. In many places with fewer resources, old methods of checking babies are not available or do not work well. Also, many health apps do not support the Sinhala language and are not accurate enough for medical use. To fix these problems, this project will create a mobile app that works in both Sinhala and English. The app will use artificial intelligence (AI) to study pictures or videos of babies and give simple, useful results. These goals will guide how the app is built, designed, and tested. The final app will be strong in science, kind in design, and easy for midwives to use in different places and languages.

Main Objectives

The main goal of this research is to create and test a mobile app that works in both Sinhala and English. This app will help midwives in Sri Lanka check babies for early signs of head shape problems and posture issues. Many midwives, especially in villages and low-resource areas, do not have easy tools to do these checks. This app will use smart computer methods called AI and computer vision to study pictures or videos of babies. It will give important health figures like CVAI (which shows if the head is flat on one side), CI (which shows the head shape), and posture angles (which show how the baby's body is positioned). The app will be built using Flutter and Firebase and will work in both Sinhala and English so that midwives with different reading levels and language backgrounds can use it easily. To make the app more **interactive and supportive**, the **output will give audio and video guidance** based on the baby's assessment. The app will be tested to make sure it is accurate, easy to use, and fits well with the way midwives work. The final goal is to give midwives a kind and smart tool that helps them care for babies better and improves early health outcomes across Sri Lanka.

Specific Objectives

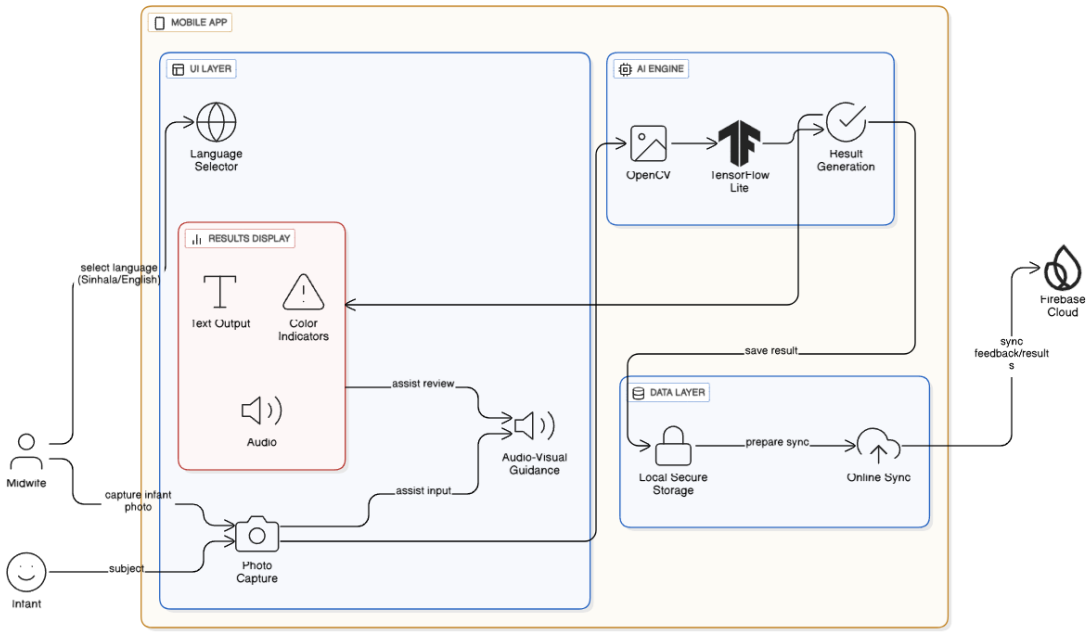
Specific objectives are small, clear steps that help guide a research project. While the main goal gives the big picture, specific objectives break it down into smaller tasks that are easier to plan and measure. In this project, the specific objectives focus on both the technical side and the user experience. The aim is to build a mobile app that uses AI to help midwives in Sri Lanka check babies for head shape and posture problems. Each objective helps make sure the app is not only medically correct but also easy to use, kind in design, and helpful for midwives in different places and languages.

1. Design and build an AI system to find head shape and posture problems from baby pictures or videos. - This step will create a smart computer model using tools called TensorFlow and OpenCV. The model will study images or videos of babies and give important health numbers like CVAI (shows if the head is flat), CI (shows head shape), and posture angles (shows how the baby's body is positioned). This will help find problems early without needing special machines.
2. Make a mobile app that works in Sinhala and English. - The app will be built using Flutter, which works on a variety of phones, and Firebase, which helps store and manage data. It will have a bilingual interface so midwives who speak Sinhala or English can use it easily.
3. Integrate audio-visual storytelling to improve understanding and interactivity. - The app will play short, pre-recorded Sinhala/English audio narrations and display simple looping motion graphics that explain the detected result and recommended action. This helps midwives, including those with lower literacy, interpret results quickly and confidently.
4. Connect the AI system to the app and only show results. - This step will put the trained AI system inside the app. It will let midwives check baby pictures or videos in real time or later. The results will be shown in a simple, color-coded

format with easy language so midwives can understand and act quickly, even without medical training.

5. Test the app with midwives to see if it is easy and friendly to use. - Midwives will try the app and give feedback. A score system called SUS (System Usability Scale) and interviews will be used to see how easy and helpful the app is. This will make sure the app feels supportive and fits well with how midwives work in both cities and villages.
6. Check how accurate and reliable the AI system is. - The AI system will be tested by comparing its results with expert opinions or trusted medical tools. Numbers like accuracy, precision, recall, and F1-score will be used to see how well the system works and if it is ready to be used in real life.

Figure 1 System Diagram



Methodology

The project will begin with a detailed literature review to study past research on mobile health applications, tools for cranial deformity and posture assessment, and bilingual health technologies. Special focus will be given to solutions that use AI for infant health screening and user interface designs that work well in low-resource settings. This review will ensure the proposed app builds on proven ideas, addresses existing gaps, and is appropriate for midwives working in rural and urban Sri Lanka.

Next, the project will focus on data collection. Publicly available cranial and posture datasets will be used first. If ethically approved, a small local dataset will be created using images or videos of Sri Lankan infants collected in homes or clinics. Parents or guardians will provide informed consent, and all images will be anonymized and securely stored. This approach ensures that the AI system learns from real-world data that reflects the conditions midwives face every day while respecting privacy and cultural norms.

After data collection, AI model development will begin. Convolutional Neural Networks (CNNs) will be built using TensorFlow and OpenCV to detect cranial abnormalities such as CVAI and CI, as well as posture issues by analyzing angles and positions. Data augmentation techniques will be applied to make the model robust under different lighting and infant positions. The trained model will then be optimized using TensorFlow Lite to run efficiently on entry-level Android devices, ensuring the solution is accessible even in low-resource settings.

Once the AI model is ready, a bilingual mobile application will be developed using Flutter so that it works on multiple platforms. Firebase will be used for cloud data storage and authentication. The user interface will be simple and intuitive, featuring culturally relevant icons and bilingual (Sinhala/English) support to make the system accessible for midwives with varying language skills and technical familiarity.

The AI module will be integrated into the mobile app so that midwives can upload images or videos and receive results in real time or after processing. Results will be displayed in a clear, color-coded format (green for normal, yellow for monitor, red for refer) along with short, easy-to-follow recommendations for action. This integration will be tested on different devices and in different environments to ensure smooth performance in rural and urban settings.

To make the system more engaging and easier to understand, audio-visual storytelling will be integrated into the app. Short pre-recorded voice narrations in Sinhala and English will explain the results, and simple looping motion graphics will visually demonstrate the detected condition and recommended next steps. This feature will help midwives, including those with limited literacy, to understand the outcomes clearly and take appropriate action confidently.

Finally, evaluation and usability testing will be conducted. The AI model's technical performance will be measured using accuracy, precision, recall, and F1 score, comparing its predictions against expert medical evaluations. Usability testing will be carried out with midwives in real-life scenarios, using the System Usability Scale (SUS) and interviews to gather feedback on ease of use, language clarity, and cultural relevance. The feedback collected will guide refinements to improve the final prototype and provide recommendations for scaling the system across Sri Lanka.

Functional Requirements

Functional requirements describe what your system must do—the core features that enable midwives to use the app effectively in real-world conditions.

1. User Authentication and Access - This ensures that only authorized users (e.g., midwives) can access the app. It protects sensitive health data and allows for future role-based access, such as adding supervisors or researchers.

2. Image and Video Capture - Midwives must be able to capture or upload images/videos of infants directly through the app. This feature is central to enabling AI analysis and must include visual guidance (e.g., framing grids) to ensure diagnostic accuracy.
3. AI-Powered Analysis - The app must process visual input using the embedded AI model to detect cranial deformities and postural irregularities. It should output clinical metrics like CVAI, CI, and posture angles, along with easy-to-understand visual indicators (e.g., color-coded risk levels).
4. Bilingual Interface - All content navigation, instructions, and results must be available in Sinhala and English. This ensures inclusiveness for midwives across linguistic backgrounds and supports training in both urban and rural contexts.
5. Result Interpretation and Guidance - AI-generated results must be presented in a simplified format, using culturally adapted language and visual cues. The app should offer actionable recommendations to support decision-making.
6. Audio-Visual Storytelling Integration – To improve understanding and interactivity, the app must play short, pre-recorded Sinhala/English audio narrations and display simple looping motion graphics that explain the detected result and recommended action. This helps midwives, including those with lower literacy, interpret results quickly and confidently.
7. Usability Feedback Collection - The app should include a built-in feedback module, allowing midwives to rate usability (e.g., SUS scores) and provide comments. This feedback loop supports iterative design and ensures the app evolves with user needs.
8. Data Privacy and Security - All personal and health data must be encrypted during storage and transmission. Users should have control over what data is stored, shared, or deleted, in compliance with Sri Lankan data protection laws.

Non-Functional Requirements

Non-functional requirements define how your system should behave - its performance, reliability, and ethical standards.

1. Performance - The app must be fast and responsive, even on entry-level Android devices. AI analysis should be completed within 10 seconds, and the interface should remain smooth during use.
2. Scalability - The system must be designed to grow, supporting more users, additional languages, and new health modules (e.g., nutrition tracking). Firebase should handle increased data volume without slowing down.
3. Usability and Accessibility - The interface must be intuitive for users with low digital literacy. This includes large buttons, simple language, culturally familiar icons, and minimal text input. Accessibility features like voice prompts or visual cues should be considered for future versions.
4. Maintainability - The codebase should follow clean architecture principles, making it easy to update, debug, and expand. Documentation should be provided for developers and health administrators.
5. Localization and Internationalization - The app must support Sinhala and English, with potential for future language additions (e.g., Tamil). Text layout, font rendering, and UI spacing should adapt to language-specific formatting and reading direction.
6. Security and Compliance - The system must comply with ethical standards and local regulations for handling health data. This includes secure authentication, encrypted storage, and audit trails for data access. Consent mechanisms must be built into any data collection process.
7. Ethical Design - AI outputs must be presented with disclaimers to avoid overdiagnosis or misinterpretation. The app should reinforce that it supports—not

replaces—clinical judgment. Emotional design principles should be applied to reduce anxiety and build trust among users.

System Requirements

System requirements define the technical specifications needed to develop, deploy, and operate a software solution effectively. They ensure that the application runs smoothly, remains secure, and performs reliably across different environments. These requirements are typically divided into two categories:

1. Hardware Requirements: The physical devices and infrastructure needed (e.g., smartphones, computers).
2. Software Requirements: The platforms, frameworks, and tools used to build and run the application.

For your project, system requirements ensure that midwives can use the app in low-resource settings and that developers have the right tools to build and maintain it.

Hardware Requirements

1. Mobile Devices - Android 8.0 or higher, minimum 2GB RAM, 16GB storage → For midwives to run the app in field conditions.
2. Development Machine - Windows/macOS/Linux, 8GB RAM, 256GB SSD, optional GPU → For AI model training, app development, and testing.
3. Test Devices - Multiple entry-level Android smartphones → To validate usability, performance, and compatibility across real-world devices.
4. Internet Access - Wi-Fi or mobile data (3G/4G) → Required for Firebase sync, updates, and remote monitoring.

Software Requirements

1. React Native → For cross-platform mobile app development (Android/iOS) and building the bilingual interface.

2. Node.js → For backend services, handling API requests, and managing data securely.
3. SQLite → For local data storage, storing app settings, results, and temporary data on the device.
4. TensorFlow Lite → For running the trained AI model on mobile to detect cranial deformities and posture issues.
5. Firebase → For authentication, cloud storage of images/videos, database management, and syncing data.
6. Android Studio → For building, testing, debugging, and deploying the mobile application on Android devices.
7. Audacity → For recording and editing audio guidance to support interactive output.
8. Adobe After Effects → For creating looping motion graphics that visually explain results and recommended actions.
9. Figma → For designing and prototyping the bilingual user interface.

Table 2 Technologies Used

Technology	**Purpose**	**Platform/Type**
React Native	Build cross-platform mobile app and bilingual interface	Mobile (Android/iOS)
Node.js	Backend services, API handling, data management	Server-side / Web
SQLite	Local storage for app settings, results, and temporary data	Mobile (Embedded Database)
TensorFlow Lite	Run AI model on device for cranial and posture detection	Mobile (On-device AI)

Firebase	Authentication, cloud storage, database management, data syncing	Cloud / Backend-as-a-Service
Android Studio	Build, test, debug, and deploy the app	Development Environment (IDE)
Audacity	Record and edit audio guidance to support interactive output	Audio Editing / Desktop Tool
Adobe After Effects	Create looping motion graphics that visually explain results and actions	Animation / Desktop Tool
Figma	Design and prototype the bilingual user interface	Design / Prototyping Tool

Figure 2 Gantt Chart

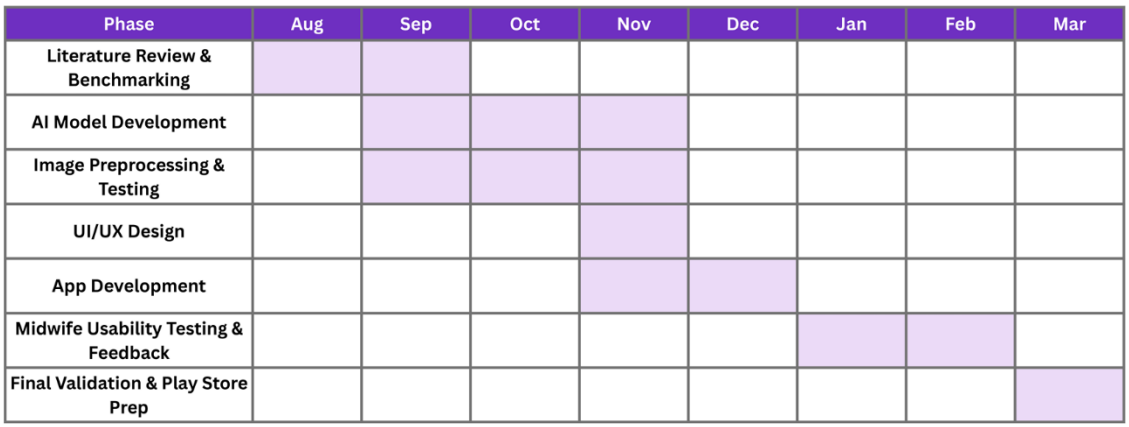
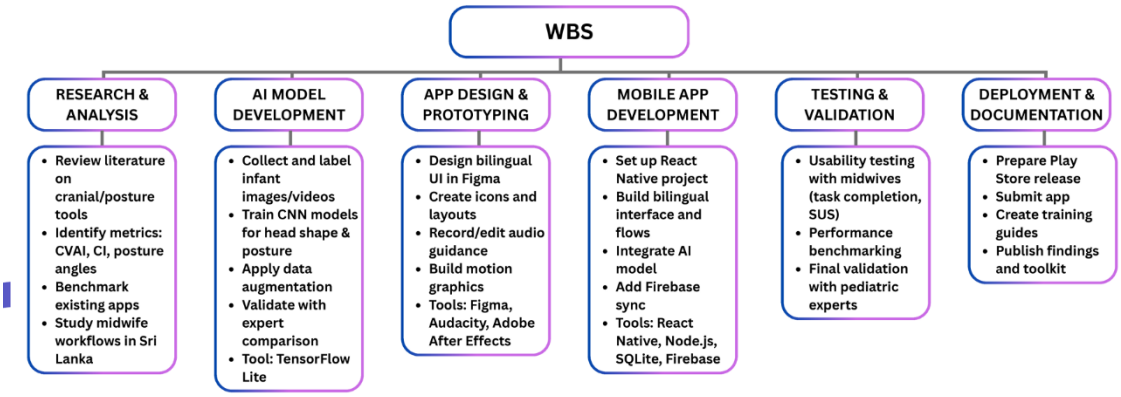


Figure 2 Work Breakdown Structure (WBS)



Description of Personal and Facilities

Supervisor

Our supervisor, Mr. Didula Thanaweera Araghchi, and Co-Supervisor Mrs. Narmada Gamage, who are experts in the technical field, will guide software development and technology integration.

External Supervisor

Our external supervisor is a midwife who has been in the field for over 26 years. She shared real experiences, how she managed mothers before, during, and after birth. Her knowledge helped us understand the real problems midwives face daily. With her guidance, we were able to plan the app feature more practically and usefully.

Facilities

The project is carried out with support from the Humanities and Nursing Departments of SLIIT. Access is provided to medical books, maternal health guidelines, and computer labs for app development and testing. This setup ensures that the mobile app is developed in line with real-world midwifery practices in Sri Lanka.

Device Access

The application will be tested on widely available Android smartphones, focusing on both entry-level and mid-range devices to ensure accessibility across different user groups. Testing will cover mothers, caregivers, and healthcare workers, ensuring smooth performance regardless of device specifications. In addition, the app will support local language interfaces, offline access for areas with poor connectivity, and secure data storage to allow continued use in diverse environments.

Budget and Budget Justification

This project requires a modest but strategically allocated budget to support the development, testing, and evaluation of a bilingual AI-assisted mobile application for infant cranial and posture monitoring. The budget prioritizes essential tools, infrastructure, and user engagement to ensure both technical feasibility and real-world usability.

Table 3 Budget Table

Item	**Estimated Cost (LKR)**	**Purpose**
Software Tools (TensorFlow, React Native, Figma, Adobe After Effects, Audacity)	15,000	For AI development, image processing, and UI/UX prototyping
Cloud Services (Firebase)	20,000	Backend infrastructure for secure data storage, authentication, and sync
Test Devices (Entry-Level Android Phones)	40,000	For performance testing and usability validation across real-world devices
Usability Testing (Compensation, Logistics)	15,000	Incentives and travel support for midwives participating in user studies
Miscellaneous / Contingency	15,000	Buffer for unexpected costs (e.g., translation, printing, accessories)
Total Estimated Budget	105,000 LKR	

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